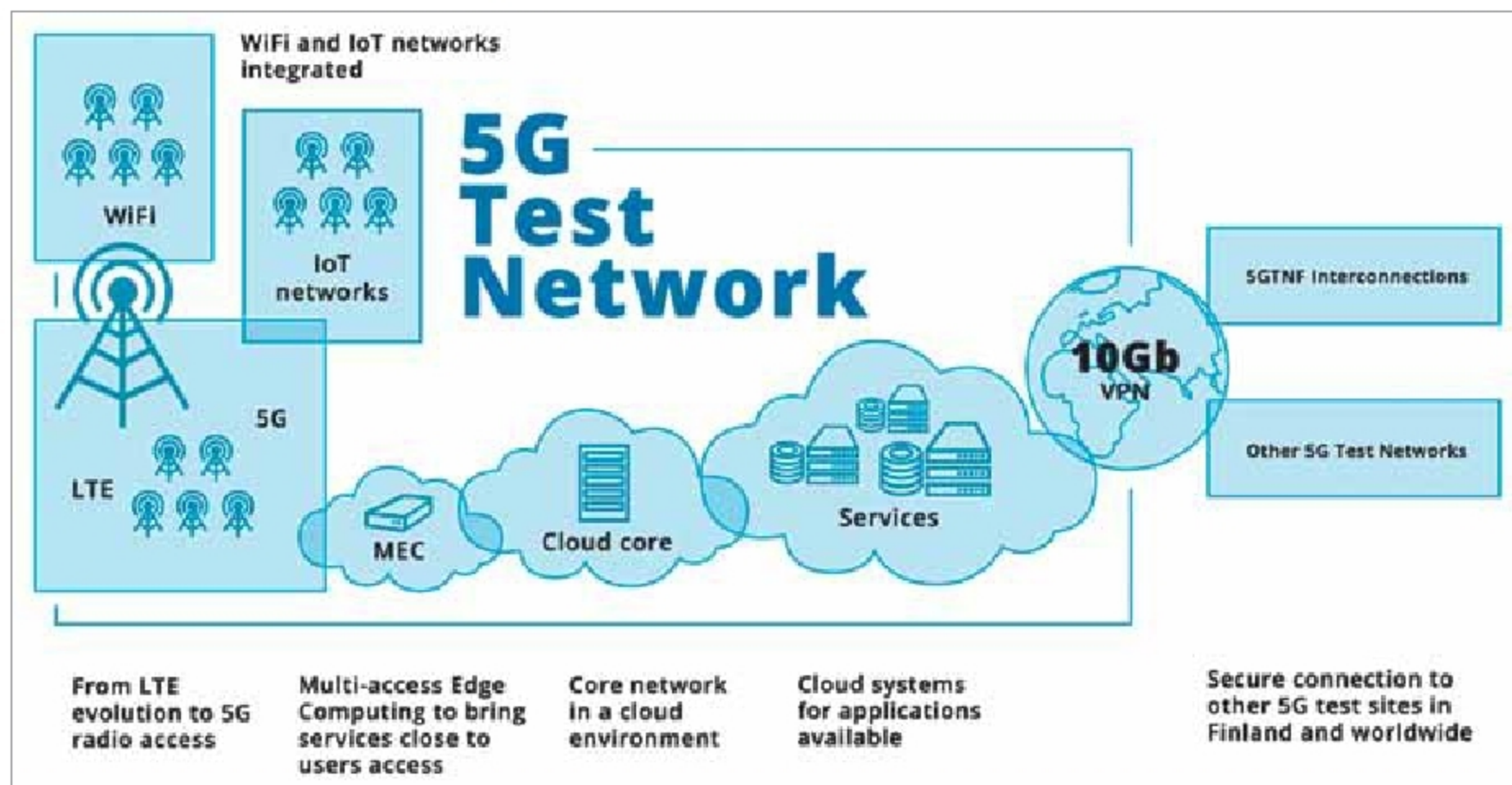




5G test network (Credit: <https://5gtn.fi/technology>)

including frequency error, time offset, cell/sector ID, modulation quality, unwanted emissions, occupied bandwidth, adjacent channel leakage ratio, transmitter spurious to 12.75GHz, and EIRP synchronisation signal block (SSB) and can be used for coverage mapping and interference hunting/testing as well.”

Out of the numerous tools available for monitoring, managing, and troubleshooting network issues in the market, one of them is WAN Killer by SolarWinds. It enables testing network performance for Wide Area Network (WAN) to ensure that it can handle large volumes of traffic when the need arises. This helps in load balancing by rerouting traffic to the available servers to prevent downtime.

For security checks, take the example of the cloud-based network scanning tool, Intruder. There are thousands of security vulnerability checks possible with the tool, including detection of bugs necessary for debugging later, configuration weaknesses, and the like. Similarly, Acunetix detects network vulnerabilities and misconfigurations such as weak passwords, gaps in firewall and router security, among others.

5G and cloud have given rise to numerous new use cases. Data collected from all connected devices

reach the cloud these days, in contrast to the traditional infrastructure. Mobile networks are extremely complex structures. There need to be cost-effective test networks and services to support these applications. Test and measurement will play a central role in the adoption of 5G devices, networks, and services. Internet of Things (IoT) solutions provider Cavli Wireless has joined hands with Maker Village, the Indian IoT and hardware innovation hub, to launch a 5G network test lab in India in the third quarter of this year.

The department of telecommunications (DoT) has recently launched its ‘5G Hackathon’ to shortlist ideas that could be converted into workable 5G products and solutions. The government received applications from companies like Airtel, Vodafone Idea, Reliance Jio, and BSNL.

Challenges

It is not possible to accurately model the performance of a network hundred per cent. Although parameters like negligible delays are not that perceptible, it is necessary to address the challenges involved in testing and measurement of networks and make accurate decisions. The cost of production of equipment is often on the high end. While improving test capabilities such that these are

user-friendly, cost and testing time also need to remain low. Managing a specialised skill-based workforce can be quite challenging too.

Telecom Regulatory Authority of India (TRAI) suggested that DoT could impose a six-month limit for a company to conduct tests on its landline and fixed-line broadband networks before commercial launch. While rules already existed for players offering wireless services in India, these draft recommendations were floated in December. Only when a connection is provided to the employees and partners, there would be no time limit for testing.

In 5G, factors like choice of the higher frequency spectrum and dense networks present newer challenges. The management of multiple tools in itself becomes an intimidating task for IT admins.

Tripathi says, “While 5G ecosystem is expanding day by day across the world, co-existence of standalone 5G (SA) and non-standalone (NSA) networks will be a real challenge in 5G testing. Interoperability with legacy networks is another important and challenging area in 5G. IoT ecosystem is full of various technologies, and therefore testing is complex. This is due to ever-increasing and evolving standards for various wireless connecting technologies and applications.” **EFY**